

Fig. 2: An actual-size, single-side PCB for the main controller section of microcontroller-based light chaser

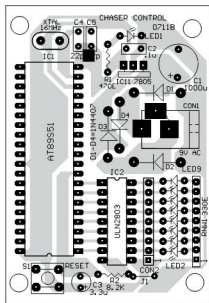


Fig. 3: Component layout for the PCB in Fig. 2

between electronic controls and power triacs to control resistive and inductive loads for a 220V AC power line. When port pin P1.0 goes high, the internal LED of MOC3020 (IC3) emits infrared light. Infrared light falls on the light-activated silicon bilateral switch and provides gate pulse to TRIAC1 to drive the electric bulb B1. Similarly, TRIAC2 through TRIAC8 drive the bulbs B2 through B8, respectively, to light-up.

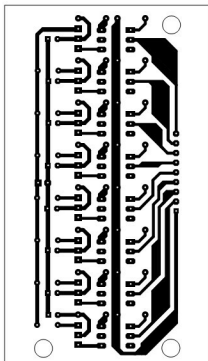


Fig. 4: An actual-size, single-side PCB for the bulb driver section of microcontroller-based light chaser

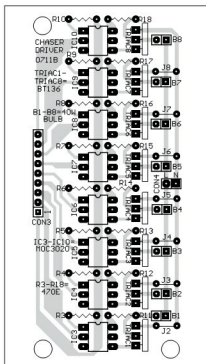


Fig. 5: Component layout for the PCB in Fig. 4

Bulbs can be connected with a two-core cable. LED2 through LED9 indicate the chasing effect on the board.

PARTS LIST

Semiconductors:

- IC1 - AT89S51 microcontroller
- IC2 - ULN2803 darlington array
- IC3-IC10 - MOC3020 optically isolated triac driver
- IC11 - 7805, 5V regulator

TRIACs:

- TRIAC1 - BT136 triac
- D1-D4 - 1N4007 rectifier diode
- LED1-LED9 - 3mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R3-R18 - 470 ohm
- R2 - 8.2-kilo-ohm
- RNW1 - 330-ohm resistor network

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 0.1 μ F ceramic disk
- C3 - 3.3 μ F, 16V electrolytic
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- X_{cr} - 12MHz crystal
- S1 - Push-to-on tactile switch
- B1-B8 - 15W, 230V bulb

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC11). Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R1 limits the current through LED1.

An actual-size, single-side PCB for the main controller section of microcontroller-based light chaser is shown in Fig. 2 and its component layout in Fig. 3. An actual-size, single-side PCB for the bulb driver section of microcontroller-based light chaser is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on the PCBs as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked errors.

Software

Chaser designer is a simple software to design various types of chaser patterns for hardware based on microcontroller 8051. Here the AT89S51 microcontroller is used, but it can be easily re-